

Chaitanya Jadhav

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SKILLS

Programming Languages	Python, JavaScript, TypeScript
Machine Learning	PyTorch, Scikit-Learn, NumPy, Pandas
LLMs / RAG / NLP	LangChain, MCP, Pydantic AI
Web & Backend	FastAPI, ASP.NET, PostgreSQL, Microsoft SQL Server
Cloud	AWS, GCP

EDUCATION

Nanyang Technological University	Aug 2021 – May 2025
Bachelor of Computing in Computer Science	GPA: 4.46/5.0

WORK EXPERIENCE

PhillipCapital Pte Ltd	Jul 2025 – Present
<i>AI Engineer</i>	

- Scaled internal AI chatbot adoption from 30 to 250 monthly active users by migrating to a multi-server MCP architecture on FastAPI, implementing role-based access control and multi-company data isolation, and launching authentication support for overseas offices.
- Replaced Turi Create with a NumPy-based item-to-item collaborative filtering system powering securities and unit trust fund recommendations for 450k+ users across 6.5M+ records in 5 global markets via the POEMS mobile app.
- Reduced customer churn model training time by 50% and inference time by 72% by parallelizing model training, while improving model performance through eliminating data leakage, refining feature engineering, and standardizing train/test preprocessing.

Panasonic R&D Center Singapore	Jan 2024 – Jul 2024
<i>AI Research Intern</i>	

- Built an internal document Q&A chatbot using RAG, Mixtral-8x7B, BAAI/bge-en-large embeddings, Ollama, and Unstructured, with configurable chunking, retrieval, and generation strategies to support systematic experimentation.
- Designed a RAG benchmarking framework for evaluating embedding models, chunking strategies, re-rankers, and answer generation models on SQuAD 2.0 and internal documents using LLM-as-a-judge for scalable evaluation.

PUBLICATIONS

C. Liu, L. Qian, **C. D. Jadhav**, J. Zhao, “Joint Optimization of Secure and Energy-Efficient Retrieval-Augmented Generation for Mobile Edge Computing,” *IEEE INFOCOM*, 2026.

C. D. Jadhav, C. Liu, J. Zhao, “Legal Retrieval-Augmented Generation with Structured Retrieval and Iterative Refinement,” *PST* 2025.

J. Q. J. Tan, **C. D. Jadhav**, J. Komar, “The Centre That Moves: A Data-Driven Perspective on Spatial Adaptations in Men’s Singles Badminton,” *ISCSS* 2025.

ACADEMIC PROJECTS

Undergraduate Research Experience on Campus Program	Aug 2023 – Jun 2024
<i>Badminton Analytics with Computer Vision and ML</i>	

- Built an end-to-end badminton analytics pipeline in Python using OpenCV and OpenPose for court isolation, video pre-processing, player detection and homography transformation onto a standardized court plane for downstream analysis.
- Benchmarked RNN, LSTM, and Conv2D models for classifying shots as forehand, backhand, or overhead, achieving 0.954 F1. Presented findings at the International Conference of Undergraduate Research 2024.

PERSONAL PROJECTS

Kinnect kinnect-sgsl.web.app

- Built a Singapore Sign Language learning and translation platform with a 1,289-sign dictionary, guided lessons, and text-to-sign translation using the OpenAI API and GCP.

Sports Event Trackers event-tracking-jkomar.pythonanywhere.com

- Built a multi-sport event-tracking web app for tennis, badminton, football, basketball, and floorball, enabling coordinate mapping, action tagging, CSV/PDF export, and keyboard-shortcut workflows for match analysis, reaching 800–2,000+ monthly visits.